

3.10 Comparative Analysis of Existing Agricultural Platforms

The following table presents a comparative analysis of existing agricultural platforms and their limitations compared to AgriTrace:

Table 3.1: Comparative Analysis of Existing Agricultural Platforms

Feature	Kisan Su-vidha	mKisan	Swachhata	PUSA Decomposer	AgriTrace
Waste Tracking	No	No	Partial	No	Yes
Collection Coordination	No	No	No	No	Yes
Payment Integration	No	No	No	No	Yes
Carbon Credits	No	No	No	No	Yes
Real-time Updates	No	Partial	Partial	No	Yes
Role-Based Access	No	No	Yes	No	Yes
GPS Location	Partial	No	Yes	No	Yes
Photo Verification	No	No	Yes	No	Yes

As the table demonstrates, AgriTrace addresses a unique combination of features that no existing platform currently provides, making it a novel contribution to the agricultural waste management domain.

3.11 Summary of Literature Review

From the literature survey, it is evident that while considerable research exists on agricultural waste management, digital platforms for waste management, and modern web technologies, there is a significant gap in integrated solutions that combine waste tracking, marketplace functionality, payment processing, carbon credit quantification, and gamification. AgriTrace is positioned to fill this gap by leveraging established technologies (Next.js, Firebase, Razorpay) in a novel application domain. The literature review also confirms the choice of Agile methodology, NoSQL databases, and responsive design as appropriate architectural decisions for this type of application.